# THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

## CLASS IX — SCIENCE (086) · MID-TERM / HALF-YEARLY PRACTICE PAPER

Session 2026-27 · School Examination Pattern

| Class: 9 | Subject: Science | Set 03 | Time: 3 Hours | Max. Marks: 80 |
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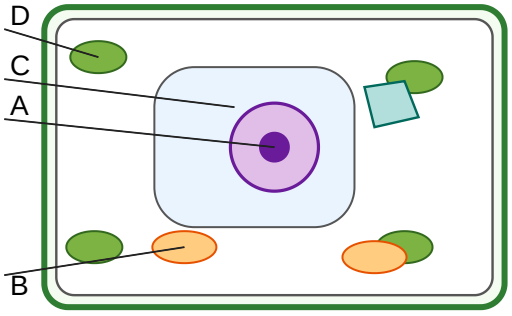
**General Instructions:**

1. This question paper consists of 39 questions in 3 sections. Section A is Biology (Q.1–16), Section B is Chemistry (Q.17–27) and Section C is Physics (Q.28–39).
2. All sections are compulsory. However, internal choice is provided in some questions. A student is expected to attempt only one of the options in such questions.
3. Section A carries 32 marks, Section B 25 marks and Section C 23 marks. Marks are printed against each question.
4. Draw neat, labelled diagrams wherever required. Use g = 10 m $s^{-2}$ wherever needed.

### SECTION A — BIOLOGY — 16 questions · 32 marks

*Q.1–7 MCQs (1 mark each); Q.8–9 Assertion–Reason. For Assertion–Reason: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.*

**1.** In the plant cell shown, which labelled structure stores cell sap and helps to keep the cell turgid? **1**




(a) A (b) B (c) C (d) D

**2.** Which of the following cell structures is found in both prokaryotic and eukaryotic cells? **1**

(a) Nuclear membrane (b) Mitochondria
(c) Ribosomes (d) Golgi apparatus

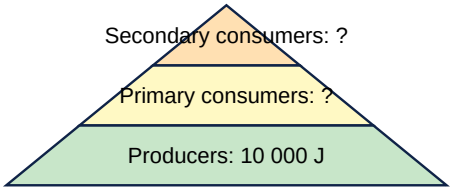
**3.** Ropes and mats are made from the fibres of coconut husk. The tissue that gives these fibres their strength is made of: **1**

(a) living cells with thin walls (b) living cells with thickened corners
(c) dead cells with lignified thick walls (d) cells with dense cytoplasm that divide

**4.** Which tissue stores fat and also acts as an insulator below the skin? **1**

(a) Areolar tissue (b) Adipose tissue (c) Cartilage (d) Tendon

**5.** In the energy pyramid shown, producers fix 10 000 J of energy. Assuming the 10 % law, the energy available to the secondary consumers is: **1**




(a) 10 J (b) 100 J (c) 1000 J (d) 10 000 J

**6.** Bryophytes (mosses) differ from pteridophytes (ferns) because bryophytes: **1**

(a) have well-developed vascular tissue
(b) lack vascular tissue and grow in moist places
(c) produce seeds
(d) bear flowers

**7.** In a flowering plant the male gamete is present in the: **1**

(a) ovule (b) ovary (c) stigma (d) pollen grain

**8.** **Assertion (A):** Skeletal muscle is called voluntary muscle. **1**
**Reason (R):** Its cells are striated and multinucleate.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

**9.** **Assertion (A):** The amount of carbon dioxide in the atmosphere would remain nearly constant if photosynthesis and respiration/decay were balanced. **1**
**Reason (R):** Plants remove $CO_2$ by photosynthesis while respiration and decomposition return it.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

**10.** Differentiate between xylem and phloem on the basis of the material they transport and the type of cells. **2**

**OR**

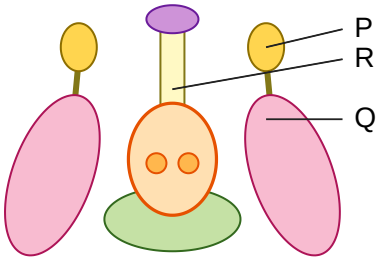
Name two permanent simple tissues of plants and write one function of each.

**11.** Define insolation. A solar panel of area 3 $m^2$ receives insolation of 600 W $m^{-2}$ for 4 hours. Calculate the solar energy received (in kWh). **3**

**OR**

What is the ozone layer and where is it present? State one cause of its depletion and one harmful effect of the depletion.

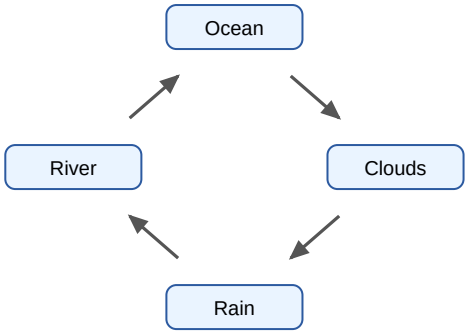
**12.** The figure shows a flower with parts P, Q and R labelled. Name them and state the function of P. **3**




**13.** Complete the table: **3**

| Organism | Kingdom |
|---|---|
| *Paramecium* | (a) _____ |
| Bread mould (*Rhizopus*) | (b) _____ |
| Fern | (c) _____ |

**14.** The figure shows the movement of water among the spheres of the Earth. Name the process by which water moves (i) from the ocean to the atmosphere, (ii) from clouds to the land and (iii) from the land back to the ocean. **3**




**15.** In a school garden, Meena observed that a young mango sapling bends in strong monsoon winds but does not break, while the trunk of an old tree is hard and rigid.

(a) Name the tissue that makes the sapling's stem flexible. 1

(b) Why is sclerenchyma rigid? 1

(c) Predict what would happen to the sapling in strong winds if its stem had only sclerenchyma instead of collenchyma. Explain. 2

**OR**

(c) Write two differences between parenchyma and sclerenchyma.

**16.** (i) Differentiate between pollination and fertilisation. **5**
(ii) Describe how fertilisation takes place in a flowering plant.

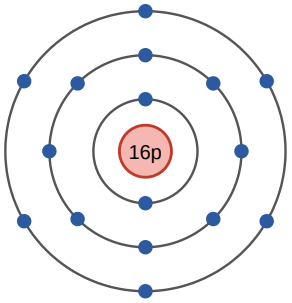
**OR**

Describe any three methods of vegetative propagation, giving one example of each. State two advantages of vegetative propagation.

## SECTION B — CHEMISTRY — 11 questions · 25 marks

*Q.17–19 MCQs; Q.20 Assertion–Reason (options as in Section A).*

**17.** The atom shown has 16 protons. Its valency is: **1**




(a) 6 (b) 2 (c) 8 (d) 16

**18.** The correct chemical formula of aluminium oxide is: **1**

(a) AlO (b) $Al_2O_3$ (c) $Al_3O_2$ (d) $AlO_3$

**19.** Light takes about how much time to reach the Earth from the Sun, if the distance is $1.5 \times 10^{11}$ m and the speed of light is $3 \times 10^{8}$ m $s^{-1}$? **1**

(a) 50 s (b) 500 s (c) 5000 s (d) 5 s

**20.** **Assertion (A):** The mass of a sealed flask does not change when a chemical reaction takes place inside it. **1**

**Reason (R):** Mass is neither created nor destroyed in a chemical reaction.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

**21.** A student adds chalk powder to water and shines a torch through it. State one observation, name the type of mixture formed and name a method to separate the chalk. **2**

**22.** Write the chemical formula of (i) magnesium chloride and (ii) the chemical name of $Na_2SO_4$. **2**

**23.** Write the number of protons, neutrons and electrons in a sodium ion, $^{23}_{11}Na^{+}$. **2**

**24.** 40 g of a 15 % (m/m) sugar solution is diluted by adding 20 g of water. Calculate the new % m/m concentration. **3**

**25.** Name the sub-atomic particle discovered by Chadwick. State its charge, relative mass and location in the atom. **3**

**OR**

State any three properties of cathode rays.

**26.** The table gives the atomic number and electronic configuration of three elements X, Y and Z.

| Element | Atomic number | Electronic configuration |
|---|---|---|
| X | 11 | 2, 8, 1 |
| Y | 17 | 2, 8, 7 |
| Z | 18 | 2, 8, 8 |

(a) Which element is chemically inert? 1

(b) What is the valency of Y? 1

(c) Write the formula of the compound formed between X and Y and state its nature. 2

**OR**

(c) Write the symbol (with charge) of the ion formed by X and state the number of protons and electrons in it.

**27.** Describe an activity to show that black ink is a mixture, using paper chromatography. Write the procedure and the observation. **5**

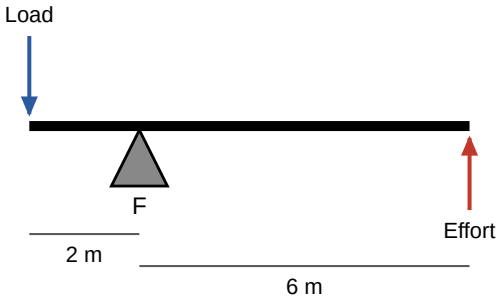
**OR**

(a) Give two differences between a mixture and a compound. (b) Classify as homogeneous or heterogeneous: air, sea water, soil, brass.

## SECTION C — PHYSICS — 12 questions · 23 marks

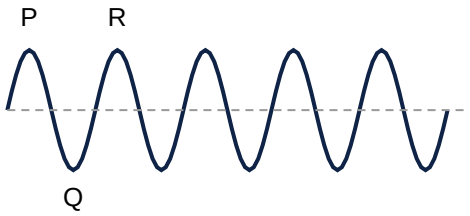
*Q.28–33 MCQs; Q.34 Assertion–Reason (options as in Section A).*

**28.** The lever shown has a load arm of 2 m and an effort arm of 6 m. Its mechanical advantage is: **1**




(a) 1/3 (b) 3 (c) 4 (d) 8

**29.** In the wave shown, P and R are consecutive crests and Q is a trough. The distance between P and R is the: **1**




(a) amplitude (b) frequency (c) time period (d) wavelength

**30.** A ball is thrown vertically upwards. At its highest point its velocity and acceleration are respectively: **1**

(a) zero and 10 m $s^{-2}$ downward (b) zero and zero

(c) 10 m $s^{-1}$ and zero (d) 10 m $s^{-1}$ upward and 10 m $s^{-2}$ downward

**31.** Two blocks of masses 2 kg and 4 kg are pushed by the same force of 12 N on a smooth surface. The ratio of their accelerations is: **1**

(a) 1 : 2 (b) 2 : 1 (c) 1 : 1 (d) 4 : 1

**32.** A machine does 3600 J of work in 1 minute. Its power is: **1**

(a) 60 W (b) 3600 W (c) 216 000 W (d) 1 W

**33.** Which of the following frequencies lies in the infrasonic range? **1**

(a) 10 Hz (b) 2000 Hz (c) 25 kHz (d) 50 kHz

**34.** **Assertion (A):** A passenger tends to fall forward when a moving bus stops suddenly. **1**
**Reason (R):** The lower part of the body stops with the bus while the upper part continues to move due to inertia of motion.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

**35.** A bus increases its speed from 18 km $h^{-1}$ to 54 km $h^{-1}$ in 5 s. Find (i) its acceleration and (ii) the distance covered in this time. **2**

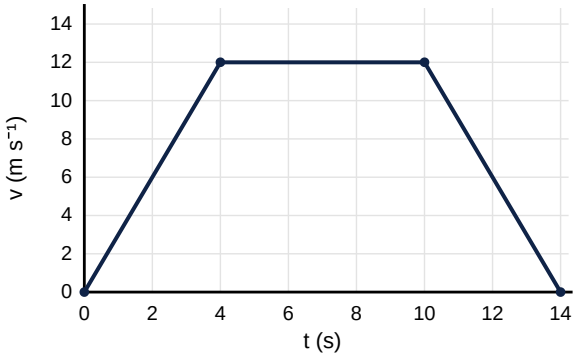
**36.** Why can sound not travel through a vacuum? **2**

**OR**

Differentiate between loudness and pitch of a sound.

**37.** A 2 kg ball is released from a height of 20 m ($g = 10\ m\ s^{-2}$). Find (i) its potential energy at the top, (ii) its kinetic energy just before it hits the ground and (iii) its speed at that instant. **3**

**38.** Asha recorded the velocity of her scooter on a straight road. The velocity–time graph is shown.




(a) Describe the motion of the scooter between t = 4 s and t = 10 s. 1

(b) Find the acceleration during the first 4 s. 1

(c) Calculate the retardation during the last 4 s. 2

**OR**

(c) Find the total distance travelled and the average speed during the 14 s.

**39.** A bullet of mass 20 g moving at $250\ m\ s^{-1}$ strikes a wooden block and stops after penetrating 0.1 m. Calculate (i) its initial kinetic energy, (ii) the work done by the resistive force, (iii) the average resistive force, (iv) the retardation and (v) the time taken to stop. **5**

**OR**

A 50 kg boy climbs a staircase of 20 steps, each 15 cm high, in 10 s. Calculate (i) the height climbed, (ii) the work done against gravity, (iii) his power, (iv) the gain in potential energy and (v) his power if he did the same in 5 s.